## Practical No. : 1

**Title:** Stop Word Removal

**Aim:** Write a program for pre-processing of a text document such as stop word removal, stemming

**Prerequisite**:Familiarity with Natural Language Processing (NLP) libraries and tools is helpful. For example, in Python, NLTK and SpaCy offer functions to easily remove stop words from text data.

**Objectives:**
● To understand implementation NLTK

**Input:** Text Data

**Outcome:**
● At end of this experiment, student will be able to illustrate the process of stop word removal and stemming

**Theory:**

Text preprocessing is a crucial step in Natural Language Processing (NLP) that involves transforming raw text data into a clean and structured format suitable for analysis. Two essential tasks in text preprocessing are **stop word removal** and **stemming**. This theory provides a conceptual understanding of these tasks and outlines a simple program to perform them.

**1. Stop Word Removal**

**Stop words** are common words that usually carry little meaningful information in text analysis. These words include articles (e.g., "the," "a," "an"), conjunctions (e.g., "and," "or"), prepositions (e.g., "in," "on"), and others. In many NLP tasks, these words are removed to focus on more meaningful words that contribute to the content or context.

**Purpose**:

- **Reduce Noise**: By removing stop words, the text becomes more concise, allowing algorithms to focus on more significant terms.
- **Improve Efficiency**: Reducing the number of words can speed up the processing time and improve the performance of text-based algorithms.

**Example**:

- Raw text: "The quick brown fox jumps over the lazy dog."
- After stop word removal: "quick brown fox jumps lazy dog"

**2. Stemming**

**Stemming** is the process of reducing words to their root or base form, known as the stem. The stem may not always be a word itself, but it represents a general concept of the word.

**Purpose**:

- **Normalize Words**: Different forms of a word (e.g., "running," "runner," "ran") are reduced to a common stem (e.g., "run"), enabling more consistent analysis.
- **Reduce Vocabulary Size**: By converting different word forms to a common stem, stemming helps reduce the dimensionality of the text data.

**Example**:

- Words: "running," "runner," "ran"
- After stemming: "run"

**Conclusion :**

Text preprocessing through stop word removal and stemming is essential for cleaning and normalizing text data, making it more suitable for various NLP tasks. By implementing these techniques, the performance of algorithms that rely on textual data can be significantly improved.